Kshitij Duraphe

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Professional Experience

Position Imaging **Stratham, NH**

Machine Learning Engineer May 2026 – Present

- Lead the company's flagship effort: **general-purpose RFID localization models that transfer to arbitrary, previously unseen environments** — an ill-posed inverse problem over sparse, multipath-corrupted RF observations with a deployment-specific forward model.
- Formulated **active sensing** as a **POMDP** with **reinforcement-learning** policies for what to measure next under partial observability; built **neural surrogates** that collapse the cost of the forward simulations behind the program.
- Own the **multimodal sensing and reasoning stack** deployed across logistics, aviation, and heavy-industrial sites, fusing camera, RFID, and Wi-Fi evidence to localize assets in metal-rich, strongly reflective spaces whose geometry differs at every installation.
- Shipped **LLM agents for scene understanding** that recover asset location when it is occluded from every camera, reasoning over surviving RF evidence and scene structure rather than degrading to manual search; in production today.
- Built **LLM agents for fleet reliability** across these deployments, correlating device, model, and network telemetry to separate site-local faults — intermittent connectivity, failing hardware, rearranged spaces — from systemic ones, and drive repair.
- Wrote custom **CUDA kernels** and GPU-level optimizations for performance-critical training and inference paths; built the edge inference stack — model compilation, graph/operator optimization, hardware-specific acceleration.
- Built the team's internal **ML experimentation platform** (ClearML, Azure, Datadog), sustaining **30+ experiments/day**, plus production **ML observability** for diagnosing deployed models.

Thespian Labs **Somerville, MA**

AI Engineer — Foundation Models and RL Post-Training Nov 2025 – Feb 2026

- Trained **Text2Motion** foundation models with **VQ-VAE** tokenization; ran multimodal pre-training over 5000+ hours of human-performance time-series on GCP.
- Implemented DARTControl-style **reinforcement-learning post-training** for controllable motion synthesis, including reward modeling and rollout infrastructure.
- Released the open-source MLOps library jobber for programmatic research-job submission to cloud GPU platforms.

Absentia Technologies **Boston, MA**

Founding Machine Learning Engineer Jan 2025 – Nov 2025

- Implemented **distributed training** with PyTorch **FSDP** (multi-GPU, mixed precision, sharding, checkpointing, fault recovery) to train vision-language models beyond single-device memory.
- Established an automated **evaluation pipeline** benchmarking model accuracy and API latency; reduced regressions **40%** and raised production stability to **99.9%**, including diagnosing a production memory leak on call.
- Architected the production platform on AWS from scratch with Terraform and Docker CI/CD, cutting deployment cycles from days to **<2 hours**; built a diffusion-based synthetic-data pipeline that reduced false positives **15%**.

Research

UniverseTBD Collaboration **Remote**

Researcher — Interpretability and Foundation-Model Evaluation Jul 2024 – Present

- First author on *The Platonic Universe* (**Spotlight**, top 1% at NeurIPS ML4PS 2025; expanded version submitted to ICLR 2027). Designed the representational-alignment **evaluation framework** over eleven model families from 10M to 10B parameters across JWST, HSC, Legacy Survey imaging and DESI spectroscopy; showed physics performance tracks **local** embedding geometry (mutual k-NN) but not **global** similarity (CKA).
- First author on two further interpretability papers (Sci-FM @ COLM 2026; NeurIPS Interpretability for Discovery 2026), establishing that model capabilities emerge in a fixed order tracking their physical difficulty, and separating layerwise where representations agree geometrically from where they agree physically.
- Creator and maintainer of platonic-universe: survey-aware loaders, reproducible eval harness, configs, CI. Run on a **6,000 GPU-hour** NCSA DeltaAI allocation (PHY250286).
- Outstanding Reviewer, ICML 2026 Mechanistic Interpretability Workshop. Invited talks at Harvard CfA (AstroAI, Jan 2026) and NeurIPS ML4PS (oral, Dec 2025).
- Primary engineer behind model architecture design, data pipelines, evals, and pretraining strategies for astronomical foundation models trained on the Multimodal Universe dataset.

Space Physics Lab, Boston University **Boston, MA**

Graduate Research Assistant Oct 2022 – May 2024

- Reconstructed corrupted sensor data with **generative inpainting** and **SwinIR super-resolution**, improving signal-to-noise for downstream models by over **60%**; M.S. thesis on deep learning over noisy all-sky imagery.
- Architected distributed ingestion (Kafka, Dask, S3) cutting processing of 3 TB+ datasets from **>24 h** to **<3 h**; built a forecasting service at **<80 ms p90** and 25 predictions/sec via async batching.

Selected Publications

1. ***K. Duraphe***, et al. *Shared Geometry Is Not Shared Physics: A Layerwise Test of the Platonic Representation Hypothesis in Astronomy.* NeurIPS Interpretability for Discovery Workshop, 2026 (arXiv:2610.04130).
2. ***K. Duraphe***, A. Kumar, S. Sourav, M. J. Smith. *What AstroPT knows about galaxies, and what that can teach us about LLMs.* Sci-FM Workshop, COLM, 2026 (arXiv:2608.22614).
3. ***K. Duraphe***, M. J. Smith, J. F. Wu, S. Sourav *(co-first). The Platonic Universe: Do Foundation Models See the Same Sky?* NeurIPS ML4PS Workshop, 2025 — **Spotlight (top 1%)**. Expanded version submitted to ICLR 2027 (arXiv:2509.19453).
4. ***K. Duraphe***, G. Bhatta, et al. *State-Dependent X-ray Variability in Cygnus X-1: A 12-Year NuSTAR Timing Study.* The Astrophysical Journal, 2026 (arXiv:2510.10746).

Open Source

platonic-universe	Foundation-model representational-alignment evaluation framework for astronomy.
rinexpy	Fast RINEX/GNSS reading and correction library; superset of `georinex`.
jobber	Declarative job submission to cloud GPU providers.
ArkOS (MIT SIPB)	DevOps and documentation for an open-source local-LLM agent platform.

Technical Skills

Languages:	Python (primary), C, C++, CUDA, TypeScript, SQL (Postgres)
ML systems:	PyTorch, FSDP, distributed training, custom CUDA kernels, GPU profiling, quantization (GPTQ, 8-bit), distillation, edge inference and model compilation, RL post-training, VQ-VAE, GNNs, VLMs
Infrastructure:	ClearML, MLflow, Slurm, Kubernetes, Docker, Terraform, Azure, AWS, GCP, Kafka, Dask, Datadog, GitHub Actions
Methods:	Evaluation design and measurement, representational analysis (CKA, mutual k-NN), Bayesian inference (MCMC), uncertainty quantification, time-series and spectral analysis, inverse problems

Education

Boston University **Boston, MA**

Master of Science with Thesis in Electrical and Computer Engineering Sep 2022 – May 2024 · GPA: 3.8/4

College of Engineering Pune **Pune, India**

Bachelor of Technology in Electrical Engineering, Minor in CS Aug 2018 – June 2022 · GPA: 3.83/4